

Status of Petrale sole stock off the U.S. West Coast in 2023



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Acronyms

NWFSC Northwest Fisheries Science Center. 5

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1 Executive Summary

1.1 Assessment Model

1.2 Reference Points, Stock Status, and Projections

The stock assessment for Petrale sole was conducted by the Northwest Fisheries Science Center (NWFSC) in U.S. West Coast. The terminal spawning biomass for the current assessment year (2022) was 8.41671. You can find more information on the results of this assessment in [Section 4.4](#). The guidelines for this report follow those outlined in [Abrams \(2014\)](#).

2 Introduction

Testing adding in an introduction for Petrale sole. There is currently no read of parameters for child documents.

2.1 Stock ID

2.2 Management History

2.3 Fishery Descriptions

2.4 Ecosystem Considerations

Ecosystem considerations and/or climate indicators were not included in this assessment.

3 Data

3.1 Life History

3.2 Catch

3.3 Indices and Standardization

3.4 Composition Data

3.5 Absolute Abundance

3.6 Environmental/Ecosystem Indicator Data

4 Assessment

4.1 Current Modeling Approach

4.2 Configuration of the Base Model

4.3 Bridging

4.4 Modeling Results

4.4.1 Parameter Estimates

4.4.2 Time Series

4.4.3 Model Fits

4.4.4 Model Diagnostics

4.5 Sensitivity Analyses

4.6 Management Benchmarks

4.7 Projections

5 Discussion

6 Acknowledgements

This document was produced using the R package asar (asar_2025?), which is free to use and publicly available on [GitHub](#).

7 References

Abrams, J.L. 2014. The effect of local fishing pressure on the size and age structure of fishes associated with rocky habitats along California's north coast. PhD thesis, Humboldt State University.

8 Tables

Table 1: Landed catch by fleet and year in (mt). (1 of 6)

Year	Landings Observed (mt) (se) - North	Landings Predicted (mt) (se) - North	Landings Observed (mt) (se) - South	Landings Predicted (mt) (se) - South
1876	0.00 (0.01)	0.00 (0.01)	1.00 (0.01)	1.00 (0.01)
1877	0.00 (0.01)	0.00 (0.01)	1.00 (0.01)	1.00 (0.01)
1878	0.00 (0.01)	0.00 (0.01)	1.00 (0.01)	1.00 (0.01)
1879	0.00 (0.01)	0.00 (0.01)	1.00 (0.01)	1.00 (0.01)
1880	0.00 (0.01)	0.00 (0.01)	11.55 (0.01)	11.55 (0.01)
1881	0.00 (0.01)	0.00 (0.01)	22.10 (0.01)	22.10 (0.01)
1882	0.00 (0.01)	0.00 (0.01)	32.65 (0.01)	32.65 (0.01)
1883	0.00 (0.01)	0.00 (0.01)	43.20 (0.01)	43.20 (0.01)
1884	0.00 (0.01)	0.00 (0.01)	53.75 (0.01)	53.75 (0.01)
1885	0.00 (0.01)	0.00 (0.01)	64.30 (0.01)	64.30 (0.01)
1886	0.00 (0.01)	0.00 (0.01)	74.85 (0.01)	74.85 (0.01)
1887	0.00 (0.01)	0.00 (0.01)	85.40 (0.01)	85.40 (0.01)
1888	0.00 (0.01)	0.00 (0.01)	95.95 (0.01)	95.95 (0.01)
1889	0.00 (0.01)	0.00 (0.01)	106.50 (0.01)	106.50 (0.01)
1890	0.00 (0.01)	0.00 (0.01)	117.05 (0.01)	117.05 (0.01)
1891	0.00 (0.01)	0.00 (0.01)	127.60 (0.01)	127.60 (0.01)
1892	0.00 (0.01)	0.00 (0.01)	138.15 (0.01)	138.15 (0.01)
1893	0.00 (0.01)	0.00 (0.01)	148.71 (0.01)	148.71 (0.01)
1894	0.00 (0.01)	0.00 (0.01)	159.26 (0.01)	159.26 (0.01)
1895	0.00 (0.01)	0.00 (0.01)	169.81 (0.01)	169.81 (0.01)
1896	0.24 (0.01)	0.24 (0.01)	180.36 (0.01)	180.36 (0.01)
1897	0.20 (0.01)	0.20 (0.01)	190.91 (0.01)	190.91 (0.01)
1898	0.15 (0.01)	0.15 (0.01)	201.46 (0.01)	201.46 (0.01)
1899	0.15 (0.01)	0.15 (0.01)	212.01 (0.01)	212.01 (0.01)
1900	0.15 (0.01)	0.15 (0.01)	222.56 (0.01)	222.56 (0.01)
1901	0.14 (0.01)	0.14 (0.01)	233.11 (0.01)	233.11 (0.01)
1902	0.14 (0.01)	0.14 (0.01)	243.66 (0.01)	243.66 (0.01)
1903	0.13 (0.01)	0.13 (0.01)	254.21 (0.01)	254.21 (0.01)

Table 2: Landed catch by fleet and year in (mt). (2 of 6)

Year	Landings Observed (mt) (se) - North	Landings Predicted (mt) (se) - North	Landings Observed (mt) (se) - South	Landings Predicted (mt) (se) - South
1904	0.13 (0.01)	0.13 (0.01)	264.76 (0.01)	264.76 (0.01)
1905	0.13 (0.01)	0.13 (0.01)	275.31 (0.01)	275.31 (0.01)
1906	0.12 (0.01)	0.12 (0.01)	285.86 (0.01)	285.86 (0.01)
1907	0.12 (0.01)	0.12 (0.01)	296.41 (0.01)	296.41 (0.01)
1908	0.11 (0.01)	0.11 (0.01)	306.96 (0.01)	306.96 (0.01)
1909	0.11 (0.01)	0.11 (0.01)	317.51 (0.01)	317.51 (0.01)
1910	0.10 (0.01)	0.10 (0.01)	328.06 (0.01)	328.06 (0.01)
1911	0.10 (0.01)	0.10 (0.01)	338.61 (0.01)	338.61 (0.01)
1912	0.10 (0.01)	0.10 (0.01)	349.16 (0.01)	349.16 (0.01)
1913	0.09 (0.01)	0.09 (0.01)	359.71 (0.01)	359.71 (0.01)
1914	0.09 (0.01)	0.09 (0.01)	370.26 (0.01)	370.26 (0.01)
1915	0.08 (0.01)	0.08 (0.01)	380.81 (0.01)	380.81 (0.01)
1916	0.08 (0.01)	0.08 (0.01)	386.42 (0.01)	386.42 (0.01)
1917	0.08 (0.01)	0.08 (0.01)	526.41 (0.01)	526.41 (0.01)
1918	0.07 (0.01)	0.07 (0.01)	423.85 (0.01)	423.85 (0.01)
1919	0.07 (0.01)	0.07 (0.01)	333.44 (0.01)	333.44 (0.01)
1920	0.06 (0.01)	0.06 (0.01)	230.49 (0.01)	230.49 (0.01)
1921	0.06 (0.01)	0.06 (0.01)	293.76 (0.01)	293.76 (0.01)
1922	0.05 (0.01)	0.05 (0.01)	424.78 (0.01)	424.78 (0.01)
1923	0.05 (0.01)	0.05 (0.01)	427.36 (0.01)	427.36 (0.01)
1924	0.05 (0.01)	0.05 (0.01)	532.86 (0.01)	532.86 (0.01)
1925	0.04 (0.01)	0.04 (0.01)	528.47 (0.01)	528.47 (0.01)
1926	0.04 (0.01)	0.04 (0.01)	521.67 (0.01)	521.67 (0.01)
1927	0.04 (0.01)	0.04 (0.01)	632.04 (0.01)	632.04 (0.01)
1928	0.00 (0.01)	0.00 (0.01)	620.09 (0.01)	620.09 (0.01)
1929	1.54 (0.01)	1.54 (0.01)	706.04 (0.01)	706.04 (0.01)
1930	1.22 (0.01)	1.22 (0.01)	658.83 (0.01)	658.83 (0.01)
1931	17.17 (0.01)	17.17 (0.01)	623.93 (0.01)	623.93 (0.01)

Table 3: Landed catch by fleet and year in (mt). (3 of 6)

Year	Landings Observed (mt) (se) - North	Landings Predicted (mt) (se) - North	Landings Observed (mt) (se) - South	Landings Predicted (mt) (se) - South
1932	41.73 (0.01)	41.73 (0.01)	556.66 (0.01)	556.66 (0.01)
1933	54.42 (0.01)	54.42 (0.01)	424.32 (0.01)	424.32 (0.01)
1934	68.97 (0.01)	68.97 (0.01)	1110.98 (0.01)	1110.98 (0.01)
1935	86.80 (0.01)	86.80 (0.01)	895.44 (0.01)	895.44 (0.01)
1936	115.33 (0.01)	115.33 (0.01)	510.98 (0.01)	510.98 (0.01)
1937	198.06 (0.01)	198.06 (0.01)	817.70 (0.01)	817.70 (0.01)
1938	134.64 (0.01)	134.64 (0.01)	989.10 (0.01)	989.10 (0.01)
1939	374.27 (0.01)	374.27 (0.01)	1160.51 (0.01)	1160.51 (0.01)
1940	602.24 (0.01)	602.24 (0.01)	714.63 (0.01)	714.63 (0.01)
1941	771.51 (0.01)	771.51 (0.01)	405.25 (0.01)	405.25 (0.01)
1942	1928.65 (0.01)	1928.65 (0.01)	277.41 (0.01)	277.41 (0.01)
1943	1959.02 (0.01)	1959.02 (0.01)	416.86 (0.01)	416.86 (0.01)
1944	1111.90 (0.01)	1111.90 (0.01)	509.50 (0.01)	509.50 (0.01)
1945	973.80 (0.01)	973.80 (0.01)	534.46 (0.01)	534.46 (0.01)
1946	1596.12 (0.01)	1596.12 (0.01)	1209.41 (0.01)	1209.41 (0.01)
1947	885.76 (0.01)	885.76 (0.01)	1336.82 (0.01)	1336.82 (0.01)
1948	1971.81 (0.01)	1971.81 (0.01)	2308.64 (0.01)	2308.64 (0.01)
1949	822.94 (0.01)	822.94 (0.01)	2246.26 (0.01)	2246.26 (0.01)
1950	1480.79 (0.01)	1480.79 (0.01)	1980.46 (0.01)	1980.46 (0.01)
1951	1084.95 (0.01)	1084.95 (0.01)	1238.49 (0.01)	1238.49 (0.01)
1952	864.21 (0.01)	864.21 (0.01)	1312.51 (0.01)	1312.51 (0.01)
1953	414.09 (0.01)	414.09 (0.01)	1514.08 (0.01)	1514.08 (0.01)
1954	634.02 (0.01)	634.02 (0.01)	1891.97 (0.01)	1891.97 (0.01)
1955	652.60 (0.01)	652.60 (0.01)	1641.72 (0.01)	1641.72 (0.01)
1956	571.36 (0.01)	571.36 (0.01)	1277.89 (0.01)	1277.89 (0.01)
1957	1051.75 (0.01)	1051.75 (0.01)	1587.99 (0.01)	1587.99 (0.01)
1958	890.87 (0.01)	890.87 (0.01)	1427.80 (0.01)	1427.80 (0.01)
1959	686.92 (0.01)	686.92 (0.01)	1194.06 (0.01)	1194.06 (0.01)

Table 4: Landed catch by fleet and year in (mt). (4 of 6)

Year	Landings Observed (mt) (se) - North	Landings Predicted (mt) (se) - North	Landings Observed (mt) (se) - South	Landings Predicted (mt) (se) - South
1960	1034.52 (0.01)	1034.52 (0.01)	1122.94 (0.01)	1122.94 (0.01)
1961	1010.41 (0.01)	1010.41 (0.01)	1538.01 (0.01)	1538.01 (0.01)
1962	1311.39 (0.01)	1311.39 (0.01)	1379.45 (0.01)	1379.45 (0.01)
1963	1205.57 (0.01)	1205.57 (0.01)	1504.99 (0.01)	1504.99 (0.01)
1964	927.02 (0.01)	927.02 (0.01)	1223.64 (0.01)	1223.64 (0.01)
1965	901.65 (0.01)	901.65 (0.01)	1207.58 (0.01)	1207.58 (0.01)
1966	961.76 (0.01)	961.76 (0.01)	1327.75 (0.01)	1327.75 (0.01)
1967	925.63 (0.01)	925.63 (0.01)	1255.79 (0.01)	1255.79 (0.01)
1968	817.32 (0.01)	817.32 (0.01)	1303.81 (0.01)	1303.81 (0.01)
1969	948.66 (0.01)	948.66 (0.01)	1300.34 (0.01)	1300.34 (0.01)
1970	1164.62 (0.01)	1164.62 (0.01)	1549.34 (0.01)	1549.34 (0.01)
1971	1314.18 (0.01)	1314.18 (0.01)	1573.50 (0.01)	1573.50 (0.01)
1972	1331.65 (0.01)	1331.65 (0.01)	1621.70 (0.01)	1621.70 (0.01)
1973	1645.58 (0.01)	1645.58 (0.01)	1304.98 (0.01)	1304.98 (0.01)
1974	2068.70 (0.01)	2068.70 (0.01)	1556.13 (0.01)	1556.13 (0.01)
1975	2006.61 (0.01)	2006.61 (0.01)	1483.24 (0.01)	1483.24 (0.01)
1976	1413.47 (0.01)	1413.47 (0.01)	1350.59 (0.01)	1350.59 (0.01)
1977	1269.72 (0.01)	1269.72 (0.01)	998.22 (0.01)	998.22 (0.01)
1978	1864.01 (0.01)	1864.01 (0.01)	1194.78 (0.01)	1194.78 (0.01)
1979	1666.00 (0.01)	1666.00 (0.01)	1388.97 (0.01)	1388.97 (0.01)
1980	1401.60 (0.01)	1401.60 (0.01)	1066.00 (0.01)	1066.00 (0.01)
1981	1236.04 (0.01)	1236.05 (0.01)	805.15 (0.01)	805.15 (0.01)
1982	1838.53 (0.01)	1838.53 (0.01)	791.80 (0.01)	791.80 (0.01)
1983	1630.29 (0.01)	1630.30 (0.01)	583.90 (0.01)	583.90 (0.01)
1984	1148.63 (0.01)	1148.63 (0.01)	590.54 (0.01)	590.54 (0.01)
1985	982.89 (0.01)	982.89 (0.01)	856.56 (0.01)	856.56 (0.01)
1986	1024.00 (0.01)	1024.00 (0.01)	726.43 (0.01)	726.43 (0.01)
1987	1381.18 (0.01)	1381.20 (0.01)	823.66 (0.01)	823.67 (0.01)

Table 5: Landed catch by fleet and year in (mt). (5 of 6)

Year	Landings Observed (mt) (se) - North	Landings Predicted (mt) (se) - North	Landings Observed (mt) (se) - South	Landings Predicted (mt) (se) - South
1988	1354.06 (0.01)	1354.07 (0.01)	795.12 (0.01)	795.12 (0.01)
1989	1312.07 (0.01)	1312.08 (0.01)	840.63 (0.01)	840.63 (0.01)
1990	1086.22 (0.01)	1086.22 (0.01)	678.42 (0.01)	678.42 (0.01)
1991	1192.57 (0.01)	1192.58 (0.01)	734.80 (0.01)	734.80 (0.01)
1992	1021.46 (0.01)	1021.46 (0.01)	532.04 (0.01)	532.04 (0.01)
1993	1039.69 (0.01)	1039.69 (0.01)	463.52 (0.01)	463.52 (0.01)
1994	825.58 (0.01)	825.58 (0.01)	549.56 (0.01)	549.56 (0.01)
1995	1066.33 (0.01)	1066.33 (0.01)	592.72 (0.01)	592.72 (0.01)
1996	1010.33 (0.01)	1010.33 (0.01)	818.27 (0.01)	818.27 (0.01)
1997	1113.93 (0.01)	1113.93 (0.01)	833.79 (0.01)	833.79 (0.01)
1998	989.95 (0.01)	989.95 (0.01)	472.82 (0.01)	472.82 (0.01)
1999	931.30 (0.01)	931.30 (0.01)	565.81 (0.01)	565.81 (0.01)
2000	1253.23 (0.01)	1253.23 (0.01)	640.04 (0.01)	640.04 (0.01)
2001	1270.22 (0.01)	1270.22 (0.01)	575.00 (0.01)	575.00 (0.01)
2002	1317.36 (0.01)	1317.36 (0.01)	479.73 (0.01)	479.73 (0.01)
2003	1661.17 (0.01)	1661.17 (0.01)	408.41 (0.01)	408.41 (0.01)
2004	1471.05 (0.01)	1471.05 (0.01)	492.92 (0.01)	492.92 (0.01)
2005	1970.53 (0.01)	1970.53 (0.01)	763.83 (0.01)	763.83 (0.01)
2006	1858.41 (0.01)	1858.41 (0.01)	751.79 (0.01)	751.79 (0.01)
2007	1333.54 (0.01)	1333.54 (0.01)	919.30 (0.01)	919.30 (0.01)
2008	1294.91 (0.01)	1294.91 (0.01)	924.80 (0.01)	924.80 (0.01)
2009	1236.74 (0.01)	1236.74 (0.01)	530.70 (0.01)	530.70 (0.01)
2010	589.72 (0.01)	589.72 (0.01)	213.38 (0.01)	213.38 (0.01)
2011	757.01 (0.01)	757.01 (0.01)	177.50 (0.01)	177.50 (0.01)
2012	896.27 (0.01)	896.27 (0.01)	221.45 (0.01)	221.45 (0.01)
2013	1776.22 (0.01)	1776.22 (0.01)	477.14 (0.01)	477.14 (0.01)
2014	1783.41 (0.01)	1783.41 (0.01)	625.33 (0.01)	625.33 (0.01)
2015	2085.62 (0.01)	2085.62 (0.01)	579.55 (0.01)	579.55 (0.01)

Table 6: Landed catch by fleet and year in (mt). (6 of 6)

Year	Landings Observed (mt) (se) - North	Landings Predicted (mt) (se) - North	Landings Observed (mt) (se) - South	Landings Predicted (mt) (se) - South
2016	2254.21 (0.01)	2254.21 (0.01)	473.42 (0.01)	473.42 (0.01)
2017	2313.91 (0.01)	2313.91 (0.01)	616.71 (0.01)	616.71 (0.01)
2018	2284.80 (0.01)	2284.80 (0.01)	609.64 (0.01)	609.64 (0.01)
2019	2079.95 (0.01)	2079.95 (0.01)	536.96 (0.01)	536.96 (0.01)
2020	1548.72 (0.01)	1548.72 (0.01)	543.41 (0.01)	543.41 (0.01)
2021	2103.03 (0.01)	2103.03 (0.01)	776.08 (0.01)	776.08 (0.01)
2022	2093.58 (0.01)	2093.58 (0.01)	966.36 (0.01)	966.36 (0.01)

9 Figures

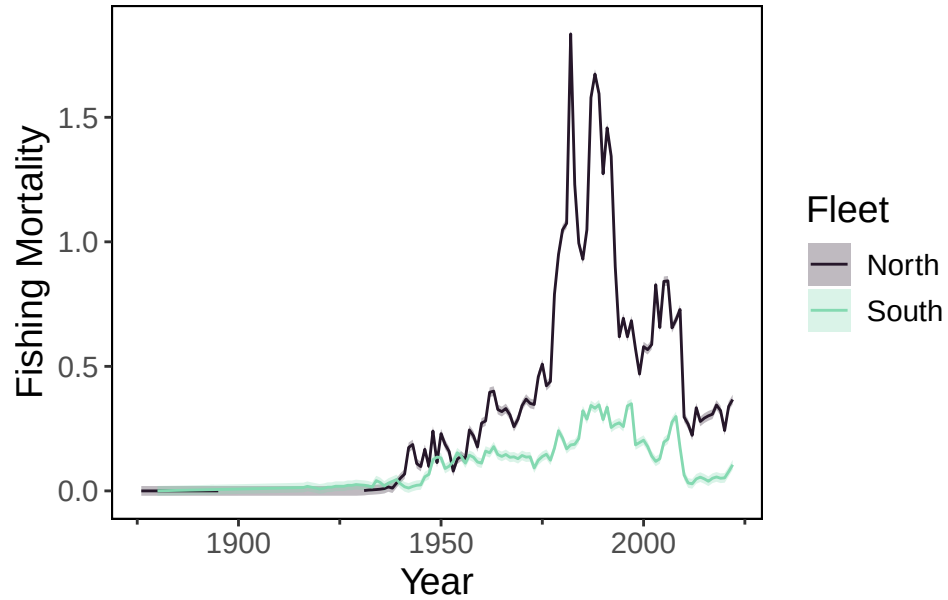


Figure 1: Fishing mortality (F) over time. The horizontal dashed line represents the target reference point (msy).

A Appendices